



ERTS Lab

Department of Computer Science and Engineering

Indian Institute of Technology Bombay, Powai

Mumbai - 400 076.



Certificate of Participation

This is to certify that **Ashutosh Bhardwaj**, a student of **Vellore Institute of Technology, Chennai, Tamil Nadu** has participated in the **e-Yantra Robotics Competition (eYRC 2025-26)**.

He/She is a member of the team having the following team members,

1. Chilukuri Venkata Chandrasa
2. Jishnu Rudraraju
3. Ashutosh Bhardwaj
4. Rama Venkata Anirudh Modiboyina

This team has participated in **KrishiCobot** theme.

Prof. Shivaram Kalyanakrishnan

Principal Investigator, e-Yantra

Associate Professor

Department of Computer Science and Engineering

Indian Institute of Technology Bombay



EYRC2025-
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[P.T.O]

eYantra

Engineering a better tomorrow

Theme: *KrishiCobot*

Theme Description:

In this theme, teams developed an autonomous smart farm management system using a UR5 robotic arm and eBot (mobile robot). The UR5 autonomously handled fertilizer containers by placing them onto the mobile robot and disposing of empty containers, while also performing vision-based inspection of harvested produce on a conveyor to identify and remove defective fruits. The mobile robot navigated the farm environment using LiDAR-based autonomous navigation, identified plant conditions using geometric structures placed, and delivered fertilizer where required. It returned to the robotic arm for container exchange. The system required coordinated integration of perception, autonomous navigation, and robotic manipulation to achieve efficient and reliable operation.

In stage 1 of the competition, teams implemented this theme in a simulator (Gazebo), and in stage 2, they got remote access to perform the task on real industrial hardware consisting of a UR5 robotic arm and eBot deployed at IIT Bombay.

Theme Learning: *Robot Operating System 2 (ROS 2), Gazebo Simulator, Computer Vision, Autonomous Navigation, LiDAR-based feature detection, Joint and Velocity-based Servoing for Manipulation, RViz 2, Multi-robot collaboration, and Algorithm Development using Python and Git.*

Certificate Level: The team has been awarded **Level 3** certificate in the e-Yantra Robotics Competition (eYRC).

Level 3 indicates: *The team showed a fair understanding of the problem statement by implementing a partial solution of the problem statement.*

Certificate Level Matrix

Certificate Level	Certificate Description
Level 1	Certificate of Merit
Level 2	Certificate of Completion
Level 3	Certificate of Participation
Level 4	Letter of Participation

* Level 1 > Level 2 > Level 3 > Level 4 (Level 1 is highest achievement)